

GESTURES GLOVE FOR GAMING

Using Arduino Uno

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The proposed gestures glove works like a mouse and keyboard in the air and allows you to control a graphic user interface (GUI). You may enjoy its novel experience while playing virtual games or its metaverse based virtual reality (VR) experience.

The glove has an MPU6050 sensor fixed on top of it, whose tilting causes an action similar to pressing a key on a computer's keyboard or movement of its mouse. You can use it to play any running game, such as Subway Surfers, which uses the A, S, D, and W keys for movement.

The author linked the sensor's movement in X-axis with the 'A' and 'D' keys and movement in Y-axis with 'W' and 'S' keys. To make it simple, he used only four keys, but with the MPU6050 sensor, you can use your creativity to do much more than that by adding more functionality to increase the fun.

The author's prototype is shown in Fig. 1 while the project's simple circuit diagram is shown in Fig. 2. The components required for the project are listed under the Bill of Material table.

As shown in Fig. 2, this simple project is built around Arduino Uno board and sensor MPU6050 only. These are connected to each other as shown in Fig. 3.

Software

Though Arduino Uno is used here, its hardware does not officially support human interface devices (HID). So, you can either use Arduino Pro Micro or Leonardo board, which officially support the HID capability.

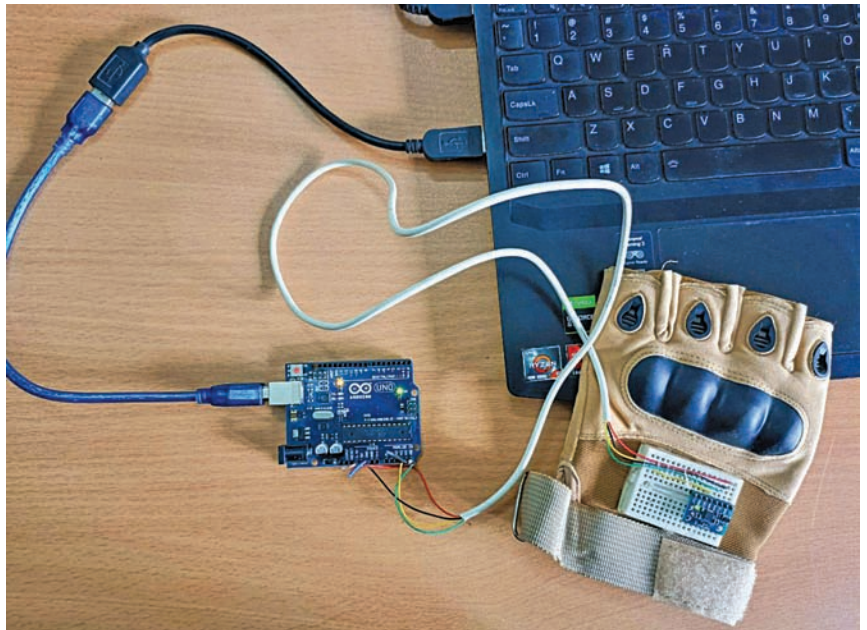


Fig. 1: Author's prototype

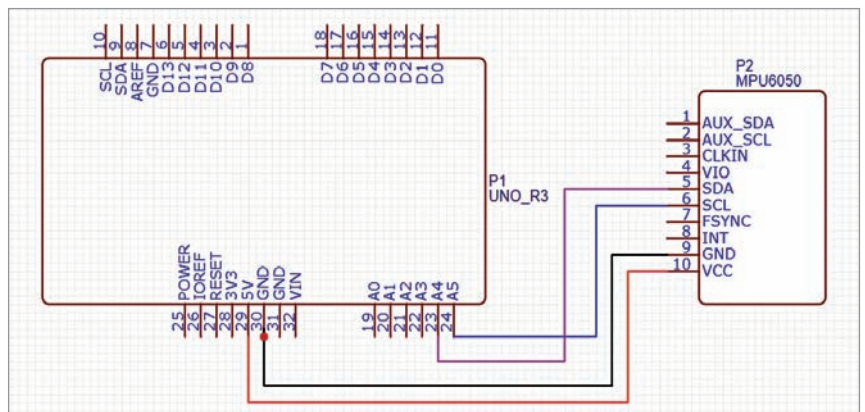


Fig. 2: Circuit diagram

BILL OF MATERIAL

Components	Quantity
Arduino Uno (P1)	1
MPU6050 sensor (P2)	1
Hand glove	1
Jumper wires	10

EFY Note

The source code of this project is available for free download at www.electronicsforu.com